3D print job: Alvik robot capstone (Nick)

Print 2 of every file. 7 STL files, 14 parts. All PLA, 0.4 mm nozzle, 0.2 mm layers, **no supports**.

Settings that matter for size

1. **Units are millimetres. Import at 100% scale.** Check the slicer shows the sizes in the table below. If a part shows about 25 times too big or too small, it was imported in inches: re-import as mm.

2. **Don't rotate the parts.** Each file already lies on the bed the way it needs, with no supports.

3. **Leave XY size compensation (horizontal expansion) at 0**, the default. The nose bracket screws into the robot and the dock's pocket fits that bracket with only 0.5 mm of play, so the sizes must come out as drawn.

4. **The dock is 196 mm long.** With its brim it needs a bed at least about 210 mm in one direction (fine on a Prusa MK3/MK4, Bambu X1/P1/A1, Ender 3; it will NOT fit an A1 mini or other 180 mm beds).

Parts

File	Qty	Size as it lies X x Y x Z mm	Colour	Settings	PLA each (about)
dock.stl	2	110 x 196 x 22	Black	4 walls, 20% infill, brim on	60 g
nose_bracket.stl	2	38 x 102 x 31	Any	4 walls, 20% infill (it is screwed on)	15 g
check_tile.stl	2	40 x 40 x 1.8	White, then black	Filament change (pause / M600) at 1.2 mm height so the raised squares print black. No colour change? Print white; the squares get coloured with a marker.	3 g
extruder.stl	2	162 x 40 x 75	Any except white	2-3 walls, 5-10% infill (lightning if available)	70-100 g
strander.stl	2	150 x 56 x 56	Any except white	2-3 walls, 5-10% infill	75-110 g
takeup_reel.stl	2	100 x 64 x 68	Any except white	2-3 walls, 5-10% infill	55-85 g
test_bay.stl	2	140 x 140 x 60	Any except white	2-3 walls, 5-10% infill	100-160 g

Total: about 1 kg of PLA for all 14 parts; the four machine models are most of it. Print the dock, nose brackets and check tiles first (about 150 g in all); the machines can follow.

Why "any colour except white": the overhead camera finds the robots partly by their white tops, so the machine models must not be white. A dark or strong colour (grey, blue, red) is best.

For Nick (not the print tech)

Parts to buy for the nose brackets, per robot: 2 Pololu #1528 mini lever switches, 4 M3 x 8 mm button-head screws, 4 M2 x 8 mm self-tapping screws, 1 Grove-to-female-jumper cable, 4 short jumper wires.

Not in this batch: the LED holder (being redesigned next week).

3D print job, page 2: AprilTag tiles (Nick, 3 Nov 2026)

8 STL files, 8 parts (one of each). PLA, 0.4 mm nozzle, 0.2 mm layers, no supports. MATTE filament if you have it (a shiny tag reflects the ceiling lights and does not read). Files: hardware/dock/stl/ in the repository.

Settings that matter

1. Millimetres, 100 % scale, no rotation: each file already lies the way it prints. XY size compensation 0.
2. TWO COLOURS by a filament change (pause / M600) at a height. The tag must come out black on white:
   - ref_corner_*.stl and ref_side_*.stl print FACE DOWN (the lips stand up): BLACK first, then change to WHITE at 0.6 mm (after layer 3). The black squares are the first three layers on the bed.
   - roof_tag_*.stl print face up, like the check tile: WHITE first, then change to BLACK at 1.2 mm.

   No colour change possible? Print all white; Nick colours the raised squares with a black marker.
3. A textured (matte) bed plate is best for the face-down tiles: their black faces come out matte.
4. Keep the first layer true to size (elephant's-foot compensation on if the slicer has it): the squares' size is the tag's size, and the cameras measure the table with it.

Parts

File	Qty	Size as it lies (mm)	Colours (order)	About
ref_corner_20.stl .. _23.stl	1 each	158 x 158 x 13.8	black, M600 at 0.6 mm, white	55 g each
ref_side_24.stl, _25.stl	1 each	155 x 158 x 13.8	black, M600 at 0.6 mm, white	50 g each
roof_tag_0.stl, roof_tag_1.stl	1 each	91 x 91 x 1.8	white, M600 at 1.2 mm, black	15 g each

Each tile's id is cut into its lip (e.g. "20 C1", "24 SIDE 1-2"). Keep them apart: which tile goes where matters.
Total about 350 g of PLA (the parts are thin: nearly solid). Print time: an estimate only, the slicer says.

For Nick (not the print tech)

Corner tiles hook the table's corners (20 = corner 1 = (0, 0), then 21, 22, 23 counter-clockwise seen from above); side tiles hook the long sides in the middle where both cameras see (24: the side from corner 1 to 2; 25: from 3 to 4). The lip goes against the table's edge; press the tile flat; tape it if it slides.
Before use: with a ruler, a tile's black square must be 120 mm (roof tiles 80 mm). Over 1 mm off: say so.
The roof tiles drop into the new roof's pocket: 91.5 x 91.5 mm, 1.2 mm deep, back-left corner cut 8 mm at 45 deg (build_ref_tiles.py prints it). The roof is being designed with the RFID team.
Paper instead: docs/hardware/reference-tags-letter.pdf (one tag a page, exact scale, cut lines).